CBSE CLASS 10 — MATHEMATICS

Top 10 Trigonometry Proof PYQs (2023–2026) | Complete Question Paper & Detailed Solutions

Part I: Question Bank

Target Score: 100%

Question 1

2023, 2024, 2026 | 3 Marks | Diff: 8/10

Prove that: $\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$

Question 2

2023, 2024 | 3 Marks | Diff: 9/10

Prove that: $\frac{\sin\theta}{1-\cos\theta} + \frac{\tan\theta}{1+\cos\theta} = \sec\theta\,\mathrm{cosec}\,\theta + \cot\theta$

Question 3

2023, 2024, 2025, 2026 | 3 Marks | Diff: 9.5/10

Prove that: $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\mathrm{cosec}\,\theta$

Question 4

2024 | 4 Marks | Diff: 8.5/10

Prove that: $\sqrt{\frac{1+\sin A}{1-\sin A}} + \sqrt{\frac{1-\sin A}{1+\sin A}} = 2\sec A$

Question 5

2025 | 3 Marks | Diff: 9/10

Prove that: $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \mathrm{cosec}\,A + \cot A$

Question 6

2025 | 3 Marks | Diff: 8.5/10

Prove that: $(\sin A + \mathrm{cosec}\,A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$

Question 7

2025 | 3 Marks | Diff: 9/10

Prove that: $\frac{1+\sec A}{\sec A} = \frac{\sin^2 A}{1-\cos A}$

Question 8

2025 | 3 Marks | Diff: 8.5/10

Prove that: $(\text{cosec } A - \sin A)(\sec A - \cos A) = \dfrac{1}{\tan A + \cot A}$

Question 9

2023, 2024, 2025, 2026 | 3 Marks | Diff: 9.5/10

Prove that: $\dfrac{1 + \tan^2 A}{1 + \cot^2 A} = \dfrac{(1 - \tan A)^2}{(1 - \cot A)^2} = \tan^2 A$

Question 10

2023 | 5 Marks | Diff: 9/10

Prove that: $\dfrac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = \dfrac{1 + \sin\theta}{\cos\theta}$